Question 9

Q: In an application like facial recognition, explain how different levels of Computer Vision are involved from image acquisition to decision making.

Theoretical Concept: CV Levels in Facial Recognition

Low-Level: Capturing the face, adjusting brightness, and filtering noise.
Mid-Level: Extracting features (edges, corners) and isolating the face from the background (segmentation/Haar Cascades).
High-Level: Using machine learning to match the extracted facial topology against a database to make a final authentication decision.

💻 OpenCV Python Solution

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```
import cv2  # Import OpenCV for image processing algorithms
import numpy as np                                            # Import Numpy for mathematical mat
import matplotlib.pyplot as plt                               # Import Matplotlib to visualize th

# Step 1: Create a synthetic image representing a human face layout
img = np.zeros((100, 100), dtype=np.uint8)  # Create a blank black image array filled with zeros
cv2.circle(img, (50, 50), 40, 255, -1) # Face base

# Step 2: Simulate Mid-Level Feature Extraction (Edge Detection)
# Feature extraction is a vital mid-level step before passing data to an AI model
edges = cv2.Canny(img, threshold1=100, threshold2=200)  # Detect edges using the Canny edge dete

# Step 3: Visualize the Mid-Level output
plt.imshow(edges, cmap='gray')  # Render the image array to the display
plt.title('Mid-Level Feature Extraction (Topology Outline)')  # Add a descriptive title to the p
plt.show()  # Show the final plot window to the user
```

🎯 Expected Output

The script displays the stark white outline of a circle against a black background. This proves how mid-level CV discards raw pixel data to provide pure topological features to high-level AI.

🚀 How to Execute & Submit

1. Google Colab Execution:

- Open [Google Colab]() and create a New Notebook.
- Click the 📋 **Copy** button on the code block above.
- Paste it into a Colab cell and press **Shift + Enter** to run.

2. Uploading to GitHub:

- In Colab, go to **File** > **Save a copy in GitHub**.
- Authorize your GitHub account.
- Select your Computer Vision Lab repository and click **OK** to commit the code.